

@NextGen_12

MINISTRY OF EDUCATION (MOE)

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)

CHEMISTRY ONLINE SIMULATED EXAM

SUBJECT: CHEMISTRY

STREAM: NATURAL SCIENCE

TIME ALLOWED: 2 Hours

GENERAL DIRECTIONS:

I. FOLLOW THE FOLLOWING RULES AND REGULATIONS IN DOING THE EXAMINATION:

- **THIS IS A COMPILED SIMULATED EXAM BASED ON THE MoEEP PLATFORM.**
- **CHOOSE ONLY THE CORRECT ANSWER FOR EACH QUESTION.**
- **USE THE PROVIDED SPACE TO WORK OUT YOUR MATHEMATICAL STEPS.**
- **ALL NECESSARY CONSTANTS ARE PROVIDED WITHIN THE QUESTIONS.**

Prepared and Compiled by: @NextGen_12

CHEMISTRY EXAMINATION

Directions: For questions 1 - 20, choose the best alternative that answers the given question.

1. The masses of different substances, liberated or deposited by the same amount of electricity, are proportional to their equivalent masses. This statement is _____.
 - a. Faraday's first law of electrolysis
 - b. the law of definite proportion
 - c. Faraday's second law of electrolysis
 - d. the law of multiple proportion
2. An unknown element has the following electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$. To which period and group do this element belongs in the periodic table, respectively?
 - a. 3 and VA
 - b. 5 and IIIA
 - c. 3 and IIIA
 - d. 4 and IIA
3. Which of the following reaction represents the chemical equation for the reaction that takes place between aluminum metal and solution of hydrochloric acid?
 - a. $2\text{AlCl}_3(\text{aq}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{Al}(\text{s}) + 6\text{HCl}(\text{aq})$
 - b. $2\text{Al}(\text{s}) + 6\text{HCl}(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2(\text{g})$
 - c. $2\text{Al} + 6\text{HCl}(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq})$
 - d. $//\text{Al} + \text{HCl} \rightarrow \text{AlCl}_3//$

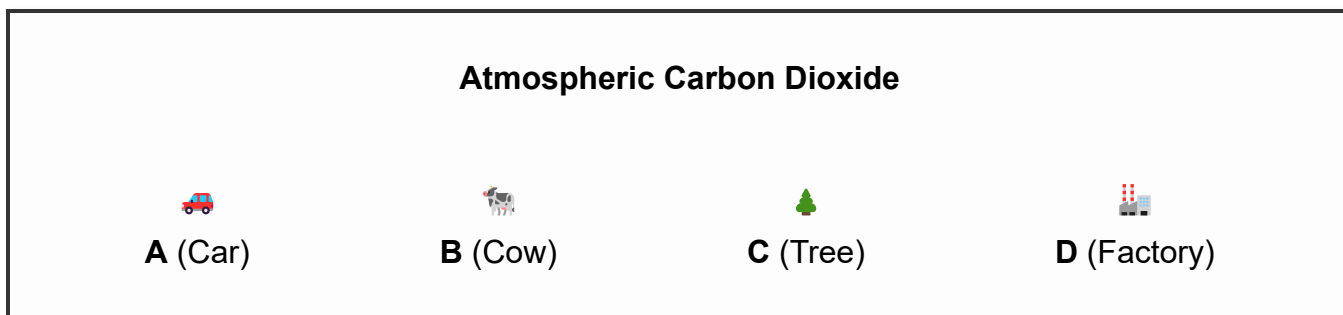
4. A cell containing AlCl_3 as an electrolyte, is placed in series with another cell containing AgNO_3 as an electrolyte. What mass of Ag is deposited, when 18 g of Al is deposited at the cathode electrode? ($M_{\text{Ag}} = 108 \text{ g/mol}$ and $M_{\text{Al}} = 27 \text{ g/mol}$)
- 36 g
 - 24 g
 - 216 g
 - 72 g
5. Nitric acid is a colorless fuming liquid with a boiling point of 83°C . However, it turns yellow as it stays for long time. Which of the following accounts for the observed property of nitric acid?
- NO_2
 - NO
 - NO_2^-
 - NO_3^-
6. Which of the following tells the exact number of atoms of different elements present in a molecule?
- Molecular mass
 - Empirical formula
 - Formula mass
 - Molecular formula
7. What is the concentration of hydrogen ion in a solution of 0.1 M nicotinic acid, $\text{HC}_6\text{H}_4\text{NO}_2$, at 25°C ? ($K_a = 1.4 \times 10^{-5}$)
- $1.2 \times 10^{-5} \text{ M}$
 - $1.2 \times 10^{-3} \text{ M}$
 - $1.4 \times 10^{-5} \text{ M}$
 - $1.4 \times 10^{-6} \text{ M}$

8. Chlorine has two naturally occurring isotopes. In a sample of chlorine, 25% of the atoms are Cl-37 and 75% of the atoms are Cl-35. What is the average atomic mass of chlorine?
- 26.25
 - 25.75
 - 35.5
 - 36.0
9. A student performs an experiment to distinguish different oxides by doing the following experiment. He/she put 100 mL KI solution in to four different test tubes. He/she acidified each of these solutions with 10 drops of dilute H_2SO_4 solution. He/she added a spatula full of CaO in the first, Al_2O_3 in the second, Na_2O_2 in the third and P_4O_{10} in the fourth test tubes. Finally, he/she added 10 mL starch solution to each of the four test tubes. In which test tube will he/she observe a color change? In the test tube that contains _____.
- Test tube 2
 - Test tube 3
 - Test tube 4
 - Test tube 1
10. Combustion of acetylene at about 3000°C with oxygen produces an intense hot flame. In which group of hydrocarbons is this compound classified?
- Alkanes
 - Aromatics
 - Alkenes
 - Alkynes

11. Phosphorus pentafluoride, PF_5 is one example of exception to the octet rule. Why does PF_5 is considered to be exceptional to the octate rule? This is because the _____.

- a. central atom has more than eight valence electrons.
- b. atoms attached to P have more than eight valence electrons.
- c. central atom has less than eight valence electrons.
- d. central atom has eight valence electrons.

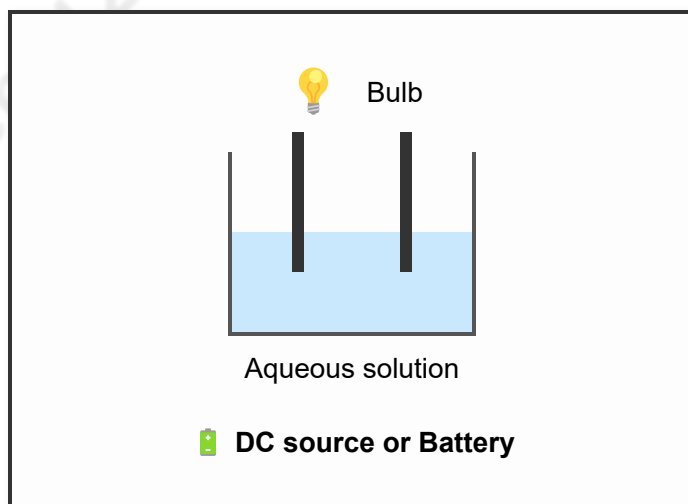
12. Consider the following diagram.



Which of the above does **NOT** contribute to global warming?

- a. B
- b. A
- c. D
- d. C

13. You are supposed to test the conductivity of the substances using a conductivity apparatus as shown in the figure below:



- Which of the following aqueous solutions does **NOT** cause the bulb to glow when the switch turns on?
- a. Sodium hydroxide solution
 - b. Copper sulphate solution
 - c. Table salt solution
 - d. Sugar solution
14. Copper metal reacts with sulfuric acid to produce copper sulphate, sulfur dioxide and water. Which of the following is the coefficient of H_2SO_4 after balancing this chemical reaction?
- a. 6
 - b. 1
 - c. 2
 - d. 3

15. What is the amount of energy emitted when an electron moves from the $n = 4$ to $n = 2$ energy level? (

$$R_H = 2.18 \times 10^{-18} \text{ J})$$

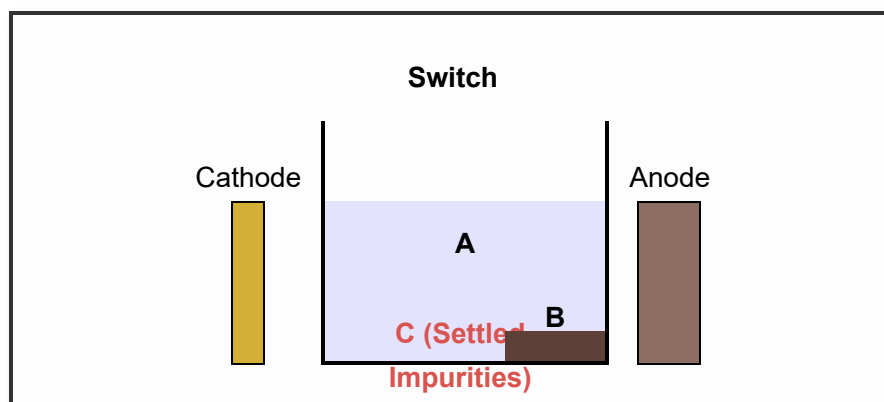
a. $-6.54 \times 10^{-18} \text{ J}$

b. $-4.09 \times 10^{-19} \text{ J}$

c. $4.09 \times 10^{-19} \text{ J}$

d. $6.54 \times 10^{-18} \text{ J}$

16. Consider the following diagram used for the electrolytic refining of copper:



What is the name of species indicated by letter 'C'?

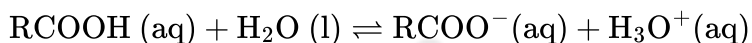
a. Copper sulphate solution

b. Strip of pure copper

c. Cathode mud

d. Anode mud

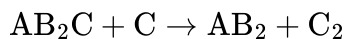
17. Consider the following equilibrium reaction of a certain weak acid, RCOOH:



Which of the following is the **CORRECT** expression of the percent ionization of RCOOH?

- a. $\frac{\text{Equilibrium concentration of H}_3\text{O}^+}{\text{Initial concentration of the acid}} \times 100$
- b. $\frac{\text{Initial concentration of the acid}}{\text{Equilibrium concentration of the acid}} \times 100$
- c. $\frac{\text{Initial concentration of RCOO}^-}{\text{Initial concentration of the acid}} \times 100$
- d. $\frac{\text{Equilibrium concentration of the acid}}{\text{Initial concentration of the acid}} \times 100$

18. Consider the reaction given below:



Which of the following successful collisions are effective in the formation of the products?

- a. The collision that results in the A and C atoms are directly facing each other.
- b. The collision that results in the C atoms are directly facing each other.
- c. The collision that results in the B and C atoms are directly facing each other.
- d. The collision that results in the A and B atoms are directly facing each other.

19. At 0 °C and 1.0 atm, the solubility of O₂ gas in water is 48.4 mole per liter. What is the molar solubility of O₂ in a saturated water solution at a partial pressure of 0.20 atm?

- a. 1.20×10^{-3} M
- b. 9.68 M
- c. 4.10×10^{-3} M
- d. 2.44 M

20. Which one of the following does **NOT** explain the chemical properties of acidic oxides? Acidic oxides react with:

- a. acids to form salt and water.
- b. basic oxides to form salts.
- c. water to form acids.
- d. base to form salt and water.

DETAILED ANSWER KEY & EXPLANATIONS (@NextGen_12)

1.

Reasoning: Faraday's Second Law of Electrolysis states that when the same quantity of electricity is passed through different electrolytes, the masses of different ions liberated at the electrodes are directly proportional to their equivalent masses (Equivalent Weights).

Answer: C

2.

Reasoning: The highest principal quantum number in the given configuration ($1s^2 2s^2 2p^6 3s^2 3p^3$) is $n = 3$, which indicates Period 3. The total number of valence electrons (electrons in the outermost shell, $3s$ and $3p$) is $2 + 3 = 5$. This places the element in Group VA (Group 15). The element is Phosphorus (P).

Answer: A

3.

Reasoning: Aluminum (Al) is an active metal that reacts with hydrochloric acid (HCl) to form aqueous aluminum chloride (AlCl_3) and release hydrogen gas (H_2). The balanced chemical equation requires 2 moles of Al and 6 moles of HCl to yield 2 moles of AlCl_3 and 3 moles of H_2 . Therefore, the correct balanced equation is $2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{AlCl}_3\text{(aq)} + 3\text{H}_2\text{(g)}$.

Answer: B

4.

Reasoning: According to Faraday's Second Law, $\frac{\text{Mass of Al}}{\text{Eq. weight of Al}} = \frac{\text{Mass of Ag}}{\text{Eq. weight of Ag}}$.

Equivalent weight of Al = $\frac{27}{3} = 9$ g/eq.

Equivalent weight of Ag = $\frac{108}{1} = 108$ g/eq.

$\frac{18}{9} = \frac{\text{Mass of Ag}}{108} \implies 2 = \frac{\text{Mass of Ag}}{108} \implies \text{Mass of Ag} = 216$ g.

Answer: C

5.

Reasoning: Pure nitric acid (HNO_3) is colorless. However, over time and especially upon exposure to light and warmth, it undergoes slight decomposition to yield nitrogen dioxide (NO_2) gas, water, and oxygen. NO_2 is a reddish-brown gas that remains dissolved in the acid, giving it a characteristic yellow or brownish tint.

Answer: A

6.

Reasoning: The **molecular formula** represents the actual and exact number of atoms of each element present in a single molecule of a compound. In contrast, the empirical formula gives only the simplest whole-number ratio.

Answer: D

7.

Reasoning: For a weak monoprotic acid, the hydrogen ion concentration can be approximated using the formula $[\text{H}^+] = \sqrt{K_a \cdot C}$, assuming ionization is very small.

$$[\text{H}^+] = \sqrt{(1.4 \times 10^{-5}) \times (0.1)}$$

$$[\text{H}^+] = \sqrt{1.4 \times 10^{-6}}$$

$$[\text{H}^+] \approx 1.18 \times 10^{-3} \text{ M} \approx 1.2 \times 10^{-3} \text{ M}.$$

Answer: B

8.

Reasoning: The average atomic mass is calculated by multiplying the relative abundance of each isotope by its atomic mass and summing the products.

$$\text{Average mass} = (0.25 \times 37) + (0.75 \times 35)$$

$$\text{Average mass} = 9.25 + 26.25 = 35.5.$$

Answer: C

9.

Reasoning: Sodium peroxide (Na_2O_2) in the third test tube is a strong oxidizing agent. In an acidic medium (H_2SO_4), it will oxidize iodide ions (I^-) from KI to form free iodine (I_2). The formed I_2 then reacts with the starch solution to produce an intense blue-black color change. The other oxides (CaO , Al_2O_3 , P_4O_{10}) do not act as oxidizing agents in this context.

Answer: B

10.

Reasoning: Acetylene, whose systematic name is ethyne (C_2H_2), contains a carbon-carbon triple bond. Hydrocarbons with one or more carbon-carbon triple bonds are classified as alkynes.

Answer: D

11.

Reasoning: In the PF_5 molecule, the central phosphorus (P) atom is covalently bonded to five fluorine atoms. Thus, it shares 10 electrons in its valence shell, exhibiting an expanded octet. The octet rule usually predicts 8 valence electrons, making this an exception.

Answer: A

12.

Reasoning: Cars (A), cows (methane emission) (B), and factories (D) are prominent sources of greenhouse gas emissions (such as CO_2 and CH_4), contributing to global warming. Trees (C), on the other hand, absorb CO_2 from the atmosphere during photosynthesis, acting as a carbon sink, and thus do NOT contribute to global warming.

Answer: D

13.

Reasoning: For a solution to conduct electricity (and thus light the bulb), it must contain free-moving ions (electrolytes). Sodium hydroxide, copper sulphate, and table salt are all strong electrolytes that dissociate into ions in water. Sugar (sucrose) is a covalent compound that dissolves as intact molecules (non-electrolyte) and does not conduct electricity.

Answer: D

14.

Reasoning: The chemical reaction of concentrated sulfuric acid with copper is a redox reaction:



Checking the balance: 1 Cu, 4 H, 2 S, 8 O on both sides. The coefficient of H_2SO_4 is 2.

Answer: C

15.

Reasoning: The energy emitted during an electronic transition in a hydrogen atom is given by the Rydberg equation:

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

The absolute amount of energy emitted is:

$$E = 2.18 \times 10^{-18} \text{ J} \times \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 2.18 \times 10^{-18} \times \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$E = 2.18 \times 10^{-18} \times \left(\frac{3}{16} \right) \approx 4.0875 \times 10^{-19} \text{ J}.$$

Rounding to proper significant figures gives $4.09 \times 10^{-19} \text{ J}$.

Answer: C

16.

Reasoning: During the electrolytic refining of copper, the impure copper is made the anode. As copper atoms oxidize and dissolve into the solution, the impurities (like Ag, Au, Pt) that are less electropositive than copper fall to the bottom of the cell beneath the anode. This accumulated material is called the "anode mud" or anode slime (labeled 'C').

Answer: D

17.

Reasoning: The percent ionization of a weak acid is defined as the ratio of the concentration of the ionized acid at equilibrium (which is equal to the equilibrium concentration of H_3O^+ or RCOO^-) to the initial concentration of the weak acid, multiplied by 100%. Therefore, $\frac{[\text{H}_3\text{O}^+]_{\text{eq}}}{[\text{RCOOH}]_{\text{initial}}} \times 100$ is the correct expression.

Answer: A

18.

Reasoning: For a chemical reaction to occur, Collision Theory states that particles must collide with sufficient energy and the correct spatial orientation. Since the reaction produces C_2 from AB_2C and a free C atom, the incoming C atom must directly strike the C atom bonded within the AB_2C molecule to form the new C – C bond.

Answer: B

19.

Reasoning: According to Henry's Law, the solubility of a gas in a liquid is directly proportional to its partial pressure ($S = k_H \cdot P$). We can set up a ratio:

$$\frac{S_1}{P_1} = \frac{S_2}{P_2}$$
$$S_2 = S_1 \times \left(\frac{P_2}{P_1} \right) = 48.4 \text{ M} \times \left(\frac{0.20 \text{ atm}}{1.0 \text{ atm}} \right) = 9.68 \text{ M}.$$

Answer: B

20.

Reasoning: Acidic oxides are oxides of nonmetals that can react with water to form acids, and react with bases (or basic oxides) to form salts. Because they already exhibit acidic character, they do **NOT** react with acids. Thus, stating that acidic oxides react with acids to form salt and water is incorrect.

Answer: A